LOFAR ANALYSIS – UNDERWATER ACOUSTIC SPECTRAL DISPLAY (LOFARgram)

Time history of the acoustic spectrum. Frequency on the horizontal axis, time on the vertical axis (waterfall display).

1 Stable narrowband tonal

A persistent vertical line at a fixed frequency indicates a stable tonal source (e.g. machinery, electrical, pump, gearbox).

2 Harmonic family

A series of vertical lines at integer multiples of a fundamental frequency ($f0$, $2f0$, $3f0$, …) indicates harmonics, typical of machinery or propeller tonals.

3 Symmetric sidebands

Pairs of vertical lines either side of a carrier frequency, spaced by a modulation frequency (fm), indicate amplitude or frequency modulation (e.g, shaft rate, blade-pass rate).

4 Slowly drifting tonal

A vertical track that gradually moves left or right with time indicates a changing frequency (e.g, manoeuvring, speed change, Doppler shift).

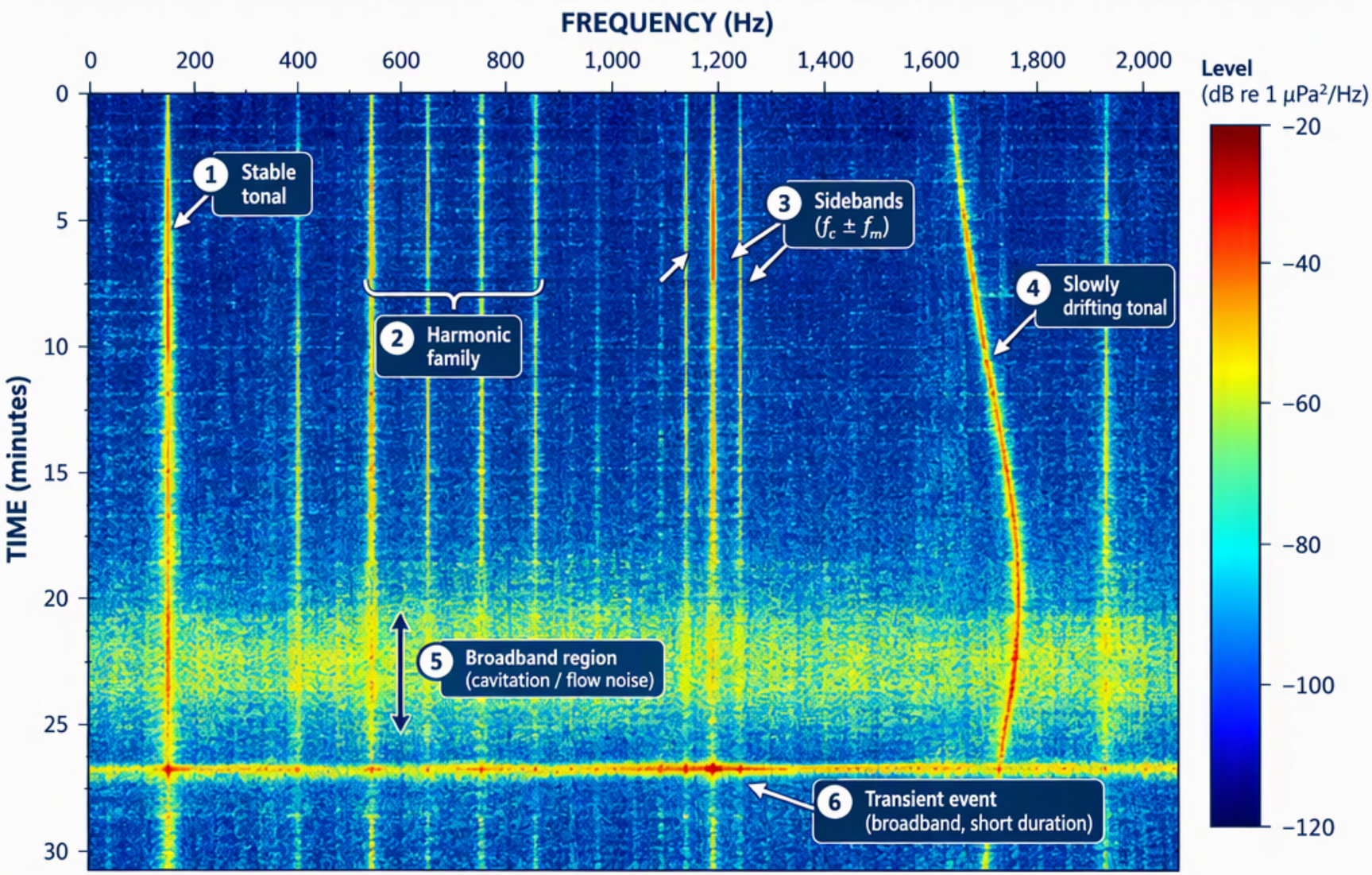
5 Broadband region

A wide band of elevated energy across a range of frequencies indicates broadband noise (e.g, cavitation, flow noise, sea state, or distant shipping).

6 Transient event

A short-duration event that appears as a horizontal streak (across many frequencies at one time) indicates an impulsive sound (e.g, sonar ping, impact, biological click, or interference).

Example LOFARgram (0 – 2 kHz, 30 minutes)



Key formulas

FFT bin spacing:

$$\Delta f = \frac{F_s}{N} = \frac{1}{T}$$

F_s = sampling frequency (Hz)
N = FFT length (samples)
T = data duration (s)

Shaft rotation rate:

$$f_s = \frac{RPM}{60}$$

Harmonic frequencies:

$$f_n = nf_0$$

n = 1, 2, 3, …

Modulation sidebands:

$$f = f_c \pm f_m$$

f_c = carrier frequency (Hz)
f_m = modulation frequency (Hz)
(e.g. shaft rate, blade-pass rate)

Display information

Frequency axis: 0 – 2,000 Hz (increasing to the right)

Time axis: 0 – 30 minutes (increasing downward)

Colour: spectral level (dB re 1 μPa²/Hz)

Display type: LOFARgram (waterfall spectrogram)

Typical marine sources

Surface ships (propellers, machinery, auxiliaries)

Biological sources (whales, dolphins)

Submarines (propellers, machinery, pumps, generators)

Ambient noise (wind, waves, rain, distant shipping)

How to interpret

1. Look for persistent vertical lines (stable tonals).
2. Identify harmonic families (integer multiples).
3. Check for symmetric sidebands (measure spacing to find modulation frequency).
4. Note any frequency drift (changing speed or manoeuvre).
5. Assess broadband energy (cavitation, flow noise, sea state).
6. Identify transient events (horizontal streaks).

Key takeaways

- Vertical lines = constant frequency (tonals, harmonics, sidebands).
- Horizontal streaks = short-duration broadband events (transients).
- Patterns reveal source type, behaviour and changes over time.
- LOFAR is a powerful tool for detection, classification and monitoring of surface ships and submarines.